Bangla Long-Form ASR and Speaker Diarization

Team Labyrinth

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BUET DL Sprint 4.0

Table of Contents

- Bangla Long-Form ASR
- Bangla Speaker Diarization

Problem & Dataset Overview

Problem Statement

- The task of transcribing **Long-Form Bangla Audio**

Dataset Overview

- **Training Set:** 113 audio files (1–2 hours each), ∼100–200 hours total
  - Paired with ground-truth text transcriptions
  - 1 corrupted file filtered out before training
- **Test Set:** 24 unannotated audio files (1–2 hours each), ∼24–48 hours total
  - No ground-truth labels; used for final inference only

Evaluation Criteria

WER Formula

$$\text{WER} = \frac{S + D + I}{N} \times 100$$

$$\text{Accuracy} = 100 - \text{WER}$$

Variables

- S = number of substitutions
- D = number of deletions
- I = number of insertions
- N = total number of words

Pitfalls

Traditional ASR Models fail on Long-form Data:

- Transformer-based architechtures scale quadratically (Limited to 30 seconds)
- Loss of temporal alignment
- Context fragmentation
- Backgrond noises and Varying pauses
- Bangla is a *low-resource* language

Proposed Approach

- Dataset preprocessing
- ASR Model Fine-tuning
- Output postprocessing

Initial Strategy: VAD (Failed)

- **Approach:** 30s audio chunks via Voice Activity Detection (VAD).
- **Method:** Approximate mapping of text annotations to chunks.
- **Result:** High temporal misalignment; model performance degraded significantly.

CTC Forced Alignment (The Fix)

- **Approach:** Word-level segmentation.
- **Method:** Extracted precise start/end timestamps for every word.
- **Result:** Clean, synchronized dataset that provided a valid "ground truth" for learning.

Word-level Forced Alignment

```
============================================================
File      : train_001.wav
Duration  : 2618.28s
Transcript: গল্পতর চ্যানেলে আপনাদের সাথে আছি আমি রাজ পড়ছিলাম হুমায়ুন আহমেদের উপন্যাস মধ্যান্য পড়ছি এর চতুর্থ ...
Words     : 5136
First 5 word alignments:
  [0.960s - 1.380s] গল্পতর
  [1.520s - 1.900s] চ্যানেলে
  [2.020s - 2.380s] আপনাদের
  [2.460s - 2.700s] সাথে
  [2.820s - 3.040s] আছি
```

Start and end timestamps for each word

```
============================================================
File      : train_002.wav
Duration  : 2542.31s
Transcript: গল্পতরু চ্যানেলে আপনাদের সাথে আছি আমি রাজ পড়ছিলাম হুমায়ুন আহমেদের উপন্যাস মধ্যান্য আজ পড়ছি এর ষষ্...
Words     : 5056
First 5 word alignments:
  [0.940s - 1.380s] গল্পতরু
  [1.460s - 1.780s] চ্যানেলে
  [1.880s - 2.200s] আপনাদের
  [2.280s - 2.500s] সাথে
  [2.640s - 2.860s] আছি
```

Start and end timestamps for each word

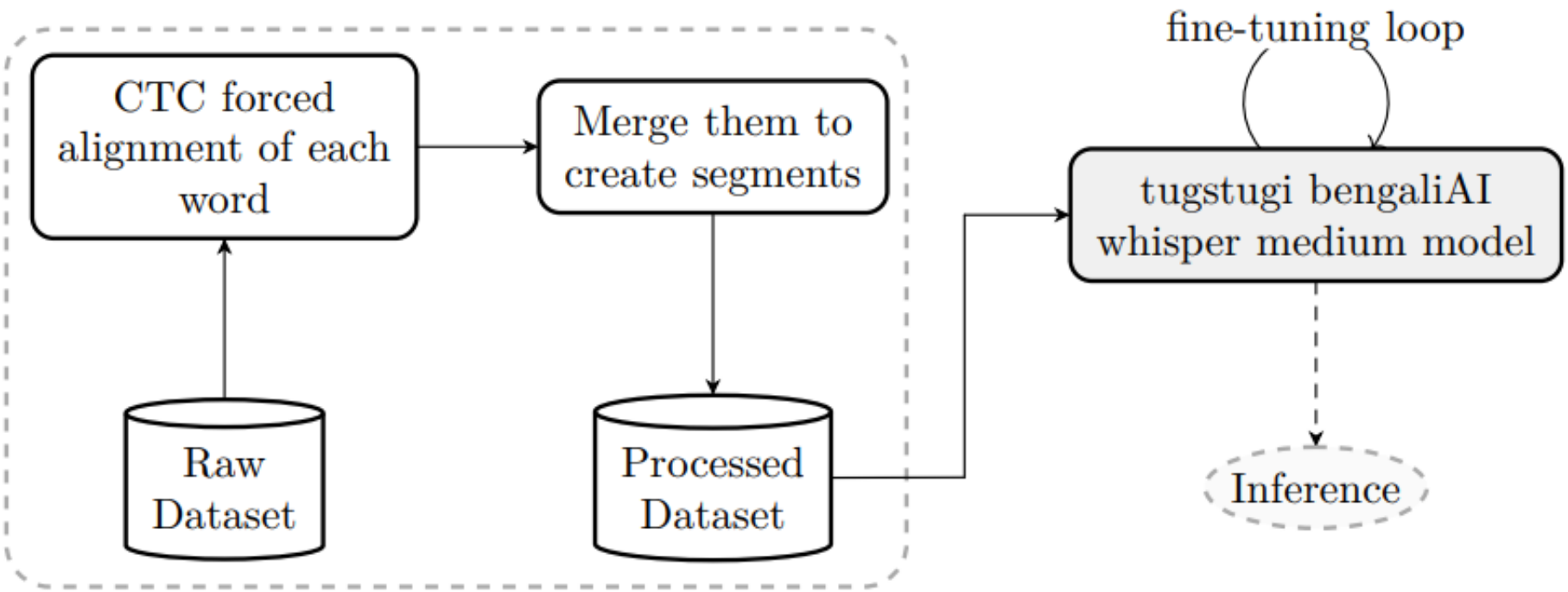
train_001.wav_chunk_0.96-28.90	গল্পতর চ্যানেলে আপনাদের সাথে আছি আমি রাজ পড়ছিলাম হুমায়ুন আহমেদের উপন্যাস মধ্যান্য পড়ছি এর চতুর্থ পর্ব রঙিলা নটিবাড়ি সোহাগঞ্জ বাজারের শেষ মাথায় মাছের আড়ত পার হয়েও আট দশ মিনিট হাঁটতে হয় রাস্তার দুপাশে আপনাতে গজিয়ে ওঠা বেশ কিছু শিমুল গাছ যে কেউ দেখে ভাববে কোন এক বৃক্ষপ্রেমী চিন্তা ভাবনা করে

Merged chunk of <30 seconds

Models Explored

Models Explored	Configuration	Public WER	Private WER
Tugstugi	**0.21988***	**0.23585***	**Fine-tuned***
Tugstugi	0.36142	0.37871	Zero-shot
Bangla-ASR	0.50047	0.54329	Fine-tuned
Mozilla Large	0.63171	0.69726	Zero-shot
Whisper Large Turbo v3	0.86594	0.88630	Zero-shot

Pipeline


fine-tuning loop
CTC forced alignment of each word
Merge them to create segments
tugstugi bengaliAI whisper medium model
Raw Dataset
Processed Dataset
Inference

Inference and Postprocessing

- Stage 1: Audio Loading
- Stage 2: Voice Activity Detection (Silero VAD)
- Stage 3: VAD-Aware Chunking
- Stage 4: Batched ASR Inference

Post-Processing

Anti-Hallucination Measures

- Repeated word removal: caps consecutive duplicates
- N-gram phrase deduplication: detects repeated 2–8 word phrases
- Excessive repetition filter: if a word exceeds 30% frequency, deduplicates via bigram filtering

Text Normalization

- Unicode NFC normalization and zero-width character removal
- Context-aware digit conversion: Bengali digits converted to Bengali words; standalone numbers preserved
- Punctuation removal and whitespace normalization

ASR Benchmarks

Model / Approach	Source	WER (%) ↓
Tugstugi (Fine-tuned - Public)	**Proposed**	**21.99**
Tugstugi (Fine-tuned - Private)	**Proposed**	**23.59**
Tugstugi (Baseline)	Existing	34.07
Tugstugi (Zero-shot - Public)	Proposed	36.14
Hishab TITU-BN	Existing	50.67

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Problem & Dataset Overview

Problem Statement

- The task of determining **"Who Spoke When?"**

Dataset Overview

- **Total:** 24 Bengali audio files, 1–2 hours each (∼25–30 hours)
- **Training:** 10 files with ground-truth CSV annotations
- **Test:** 14 unannotated files for inference

Evaluation Criteria

Diarization Error Rate (DER)

$$\text{DER} = \frac{\text{False Alarm} + \text{Missed Speech} + \text{Speaker Confusion}}{\text{Total Reference Speech}}$$

Existing Models and Limitations

Two notable models:

- **pyannote/speaker-diarization-3.1***
- NVIDIA NeMo

- Out-of-the-box models have high DER
  - Noise
  - Varying recording volumes
  - Overlaps
- Scarce annotated Bangla diarization datasets

Fine-Tuning Strategies Explored

Strategy	Public DER ↓	Private DER ↓	Key Idea
Zero-Shot (No Fine-tuning)	0.31482	0.39217	Baseline; multilingual pyannote out-of-the-box
Normal Fine-tuning (Base)	0.23147	0.31129	Trained directly on raw Bengali audio
Fine-tuning + LR Scheduling	0.24891	0.32740	Cosine annealing scheduler during training
Fine-tuning + Gradient Clipping	0.25103	0.33012	Clip gradients at 1.0 to stabilize training
Fine-tuning + Demucs Refinement	0.21621	0.33454	Source separation to isolate vocals before training
Fine-tuning + Data Augmentation	**0.21460***	**0.32663***	Dynamic Gain ($\pm$6.0 dB, $p = 0.4$) for robustness

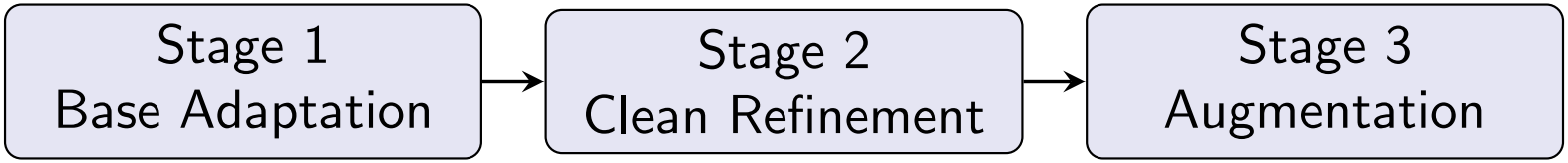
Best Strategy

Fine-tuning + Data Augmentation achieves the lowest Public DER (0.21460) and a low Private DER (0.32663)

Fine-Tuning Strategy: Diarization

Approach: Three-Stage Curriculum Learning

- Base model: `pyannote/segmentation-3.0` (multilingual speaker segmentation)
- Progressively guides model from broad acoustic adaptation to specific vocal refinement



Stage 1: Base Adaptation (Real-World Acoustics)

Problem

Multilingual models lack Bengali phonetic cadence and struggle with conversational overlap

Solution

- Adapt to Bengali phonetic cadence
- Trained on unaltered, noisy raw audio chunks
- Learns Bengali speech characteristics through natural background noise, static, and music
- Best-performing checkpoint exported for Stage 2

Stage 2: Clean Refinement (Source Separation)

Problem

Stage 1 model trained on noisy audio struggles to distinguish speaker boundaries from background noise

Solution

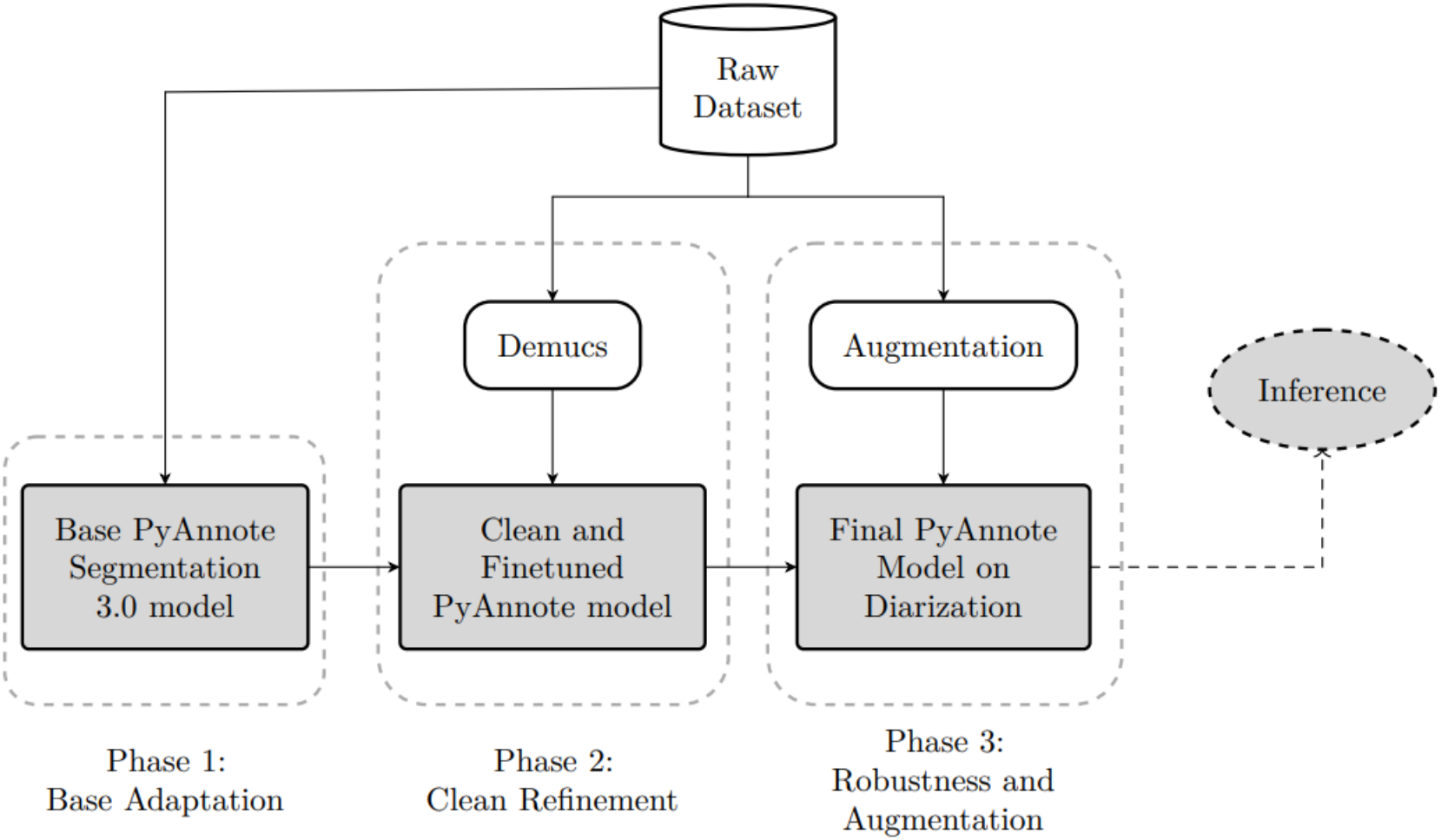
- **Goal:** Isolate distinct speaker embeddings
- **Vocal Isolation:** Demucs hybrid Transformer strips background noise
- **Audio Standardization:** Isolated vocals converted to mono 16 kHz
- **Clean Fine-Tuning:** Resume from Stage 1 weights exclusively on Demucs-cleaned audio

Stage 3 Robustness via Dynamic Augmentation

- **Problem:** Models trained on clean audio overfit to artificial cleanliness
- **Solution:** Controlled stochastic distortions via `torch_audiomentations`

Technique	Parameter	Detail
Dynamic Gain Augmentation	± 6.0 dB	Applied with 40% probability
Chunk Duration	10 seconds	Increased from Stage 1
PyAnnote Wrapper	ManualExternal	Catches augmented tensors, bundles target labels

Pipeline


Raw Dataset
Demucs
Augmentation
Inference
Base PyAnnote Segmentation 3.0 model
Clean and Finetuned PyAnnote model
Final PyAnnote Model on Diarization
Phase 1: Base Adaptation
Phase 2: Clean Refinement
Phase 3: Robustness and Augmentation

Curriculum Learning Hyperparameters

Parameter	**Phase 1**	**Phase 2**	**Phase 3 (Augmentation)**
Chunk Duration	5.0 s	5.0 s	10.0 s
Batch Size	32	32	32
Learning Rate	5×10^{-5}	5×10^{-5}	5×10^{-5}
Epochs	Variable	Variable	20
Audio Source	Raw Audio	Demucs Vocals	Augmented Audio
Augmentation Protocol	None	None	Dynamic Gain (± 6.0 dB, $p = 0.4$)
Loss Function	Cross-Entropy	Cross-Entropy	Cross-Entropy

Speaker Diarization Benchmarks

System / Training Strategy	Source	DER (%) ↓
Fine-tuning + Data Aug. (Public)	**Proposed**	**21.46**
Fine-tuning + Data Aug. (Private)	**Proposed**	**32.66**
pyannote.audio (pretrained)	Existing	40.08
Silero VAD + ECAPA + Clustering	Existing	61.50
WebRTC VAD + ECAPA + Clustering	Existing	73.71

Thank You!